DATA 3464: Fundamentals of Data Processing

Missing and weird data

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Topic overview

- What to do with missing data
- Detecting and handling outliers

Resources used:

- Feature Engineering Chapter 8
- Hands on Machine Learning with Scikit-Learn and Tensorflow/PyTorch, Chapter 2. Available at MRU Library
- Scikit-learn user guide: Chapters 2 and 7

The problem

- As you've seen, real-world data is messy
- Missing values are common, other values don't make sense
- We need to decide how to deal with these problems

> *What examples have we seen so far?*
>
> *Why might data be missing or weird*?*

*I'm using "weird" as an informal catch-all for unexpected or outlier values

Missing data

When data are missing in the *features* we have a few options:

0. Do nothing! Some algorithms (e.g. Decisions Trees) can handle missing values
1. Remove *features* with missing values
2. Remove *samples* with missing values
3. Invent a new value to represent "missingness"
4. **Impute** a value based on other data

> *Most important: understand why data are missing (more EDA!)*

Option 1: removing features

- If a feature has:
  - A high proportion of missing values, *and*
  - Little apparent relationship to the target *or*
  - its information is redundant with other features
- It may be reasonable to remove it entirely. You can drop it, e.g.:

  ```
  df.drop(columns=['feature_name'], inplace=True)
  ```

  or (probably more reliable) just not select it when building your pipeline

Option 2: removing samples

- If only a small number of samples have missing values, you can drop them **from the training data**:

```
train.dropna(subset=[["features","we","care","about"]], inplace=True)
```

- Good idea if the same samples have missing values from multiple features
- Still useful to explore *why* data are missing

> *What should we do for inference?*

Option 3: invent a new value

- Categorical features: add a new category for "missing"
- Add a new binary feature indicating whether the value was missing
- I have seen advice to use extreme values for numerical features, like the -1 income in the OKCupid dataset, but I'm not convinced this is a good idea

> *Case study: where missingness is informative*

Option 4: impute missing values

- Fill in the missing values with an "educated guess"
- Replace missing value with:
  - constant
  - mean, median, or mode (`most_frequent`)
- Use other features to infer missing value:
  - K-nearest neighbours
  - simple models to predict missing values
- Can be combined with option 3 to indicate missing features
- How much to impute? Feat.Engineering suggests no more than 20%

Choosing an imputation strategy

Strategy	When to use
Constant	When there is a reasonable default value
Mean	Numeric features with normal distribution
Median	Numeric features with extreme outliers
KNN	Relationship with other features
Missingness indicator	If missingness seems informative

Outliers

> *An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism. -- D. M. Hawkins*

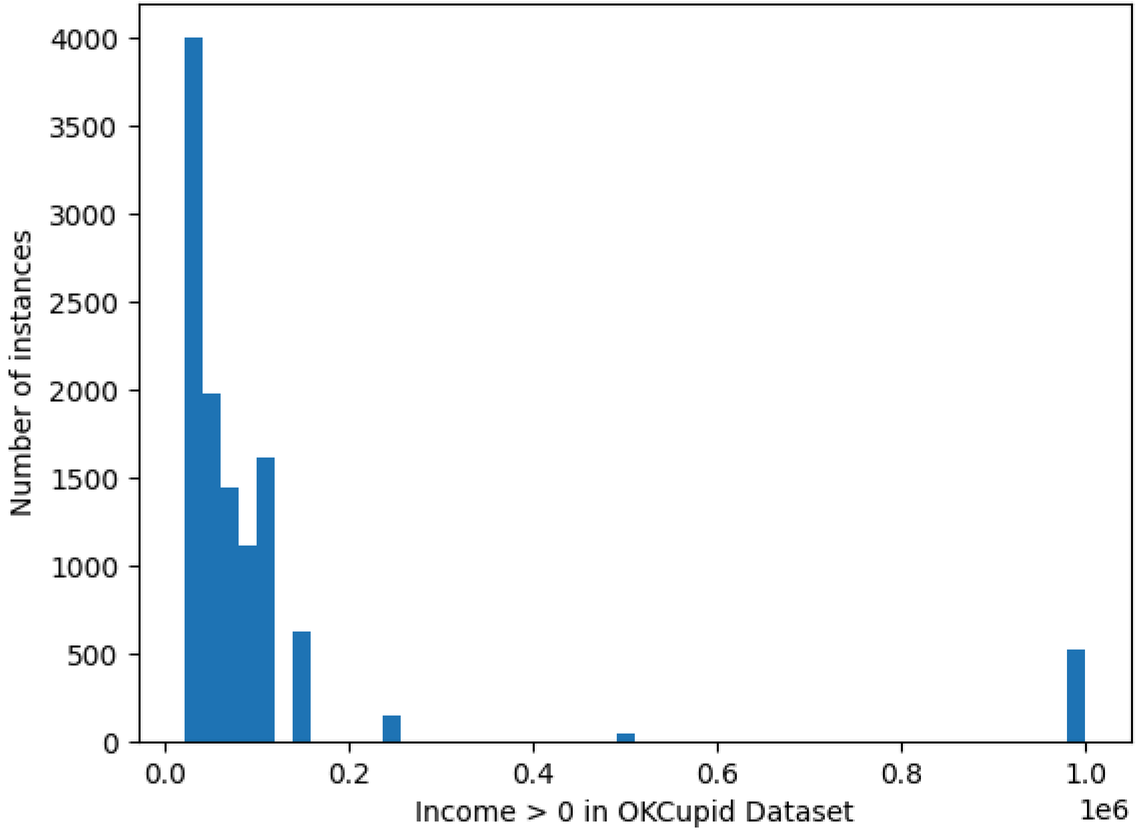
- There are many entire books dedicated to outlier detection
- Useful for anomaly detection, e.g:
  - fraud detection
  - failure prediction
  - Many other applications
- Our focus is on dealing with outliers in preprocessing

Hawkins, D. M. (1980). Identification of Outliers.

Where we left off on February 5

Detecting outliers

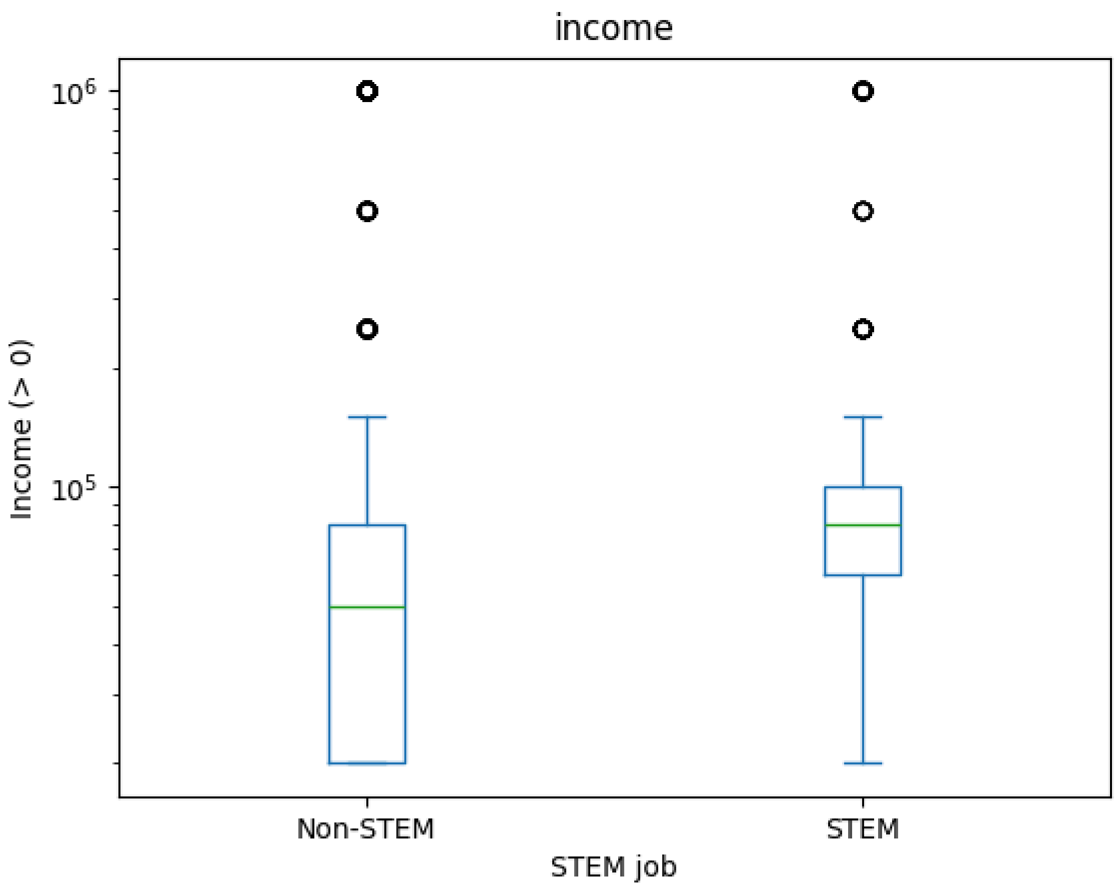
- Visually as part of EDA
- Statistically, e.g. $> 3\sigma$
- Algorithmically, e.g. Isolation Forests
- As usual, context and domain knowledge are essential



Context matters!

- Look at the relationship between outliers and target (training dataset, of course)

> *What do the dots on box plots represent?*

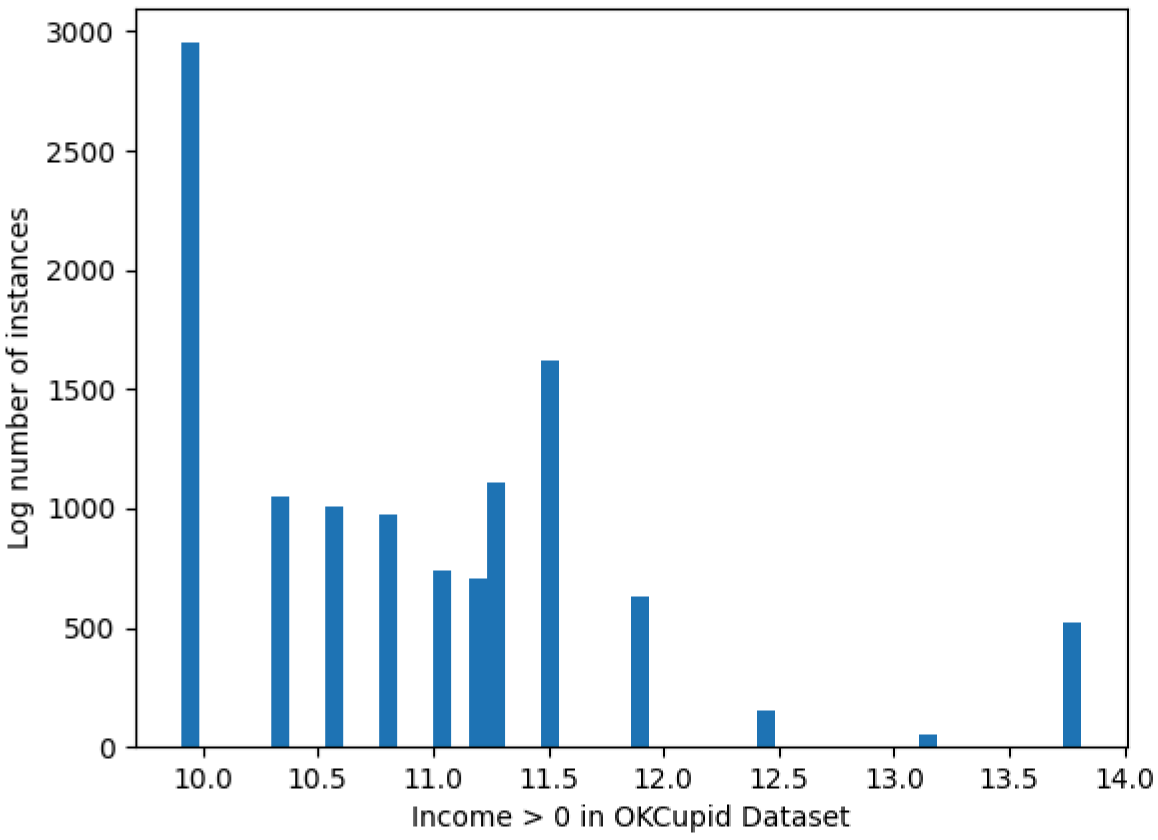


What to do about outliers?

1. Data transformations
2. Drop the samples
3. Encode them somehow
4. Leave them alone

As usual, very data- and model-dependent. Tree based methods are particularly impacted by outliers!

> *Any other ideas?*



Nonlinear transformations

- Transforming the data does not actually remove the outlier
- Can help make the relationship less extreme

Dropping the samples

- In general, not a thing I love to do
- If you drop a sample from training, you need to decide what to do at inference
- My opinion: Only do it if you're confident it's an error in the dataset

> *Can you think of an example?*

- What can you do at inference time when outliers are encountered?

Encoding outliers

A few other options that might fall into "encoding":

- Just like with missing values, binary column indicating outlier/inlier
- Remove the values and convert them to missing
- Bin or impose a cap (floor/ceiling) on the value
- Replace the numeric value with a rank or quantile bucket
- Probably other things!

Leave them alone

If your outliers are:

- Real values (not data entry or other errors)
- Representative of things that might happen during inference

Then you probably want to keep them!

> *Consider using a [RobustScalar](#) to standardize if you have lots of outliers*

Missing values and outliers in the target

- If you have missing values in the target, you probably want to avoid imputing
- This is a good case for dropping samples from **training**!
- Outliers are trickier -- again, check if they're real or mistakes
- There may be a case for transforming the target

Coming up next

- Interactions between variables
- Feature selection
- Midterm stuff
- Reading week!!!!